Apples and pears? Impacts of using the Jacobian scaled Conditional Akaike Information Criteria for model selection in linear mixed models

Advanced Small Area Estimation (Prof. Schmid)

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1 Introduction

In model-based statistical analysis, selecting the appropriate covariates is one of the most important aspects. That decision can either be drawn from theoretical considerations, but also based on metrics indicating the model quality. In case of the latter approach, most metrics require the dependent variable to be on the same scale to be comparable. The problem of variable selection also exists in context of Linear Mixed Models, defined as

$$y_i = X_i\beta + Z_i u_i + \varepsilon_i \tag{1}$$

with y being the dependent variable, $i = 1, ..., D$ indexing the respective area D as well as design matrices X and Z. β indicates a vector of coefficients, later to be estimated with the model, and u is a vector of area-level random effects. Lastly, ε is a vector of area-level errors. Importantly, u and ε are assumed to be independent and normally distributed with $u \sim N(0, G)$ and $\varepsilon \sim N(0, \sigma^2 I_{N_i})$ respectively.

One metric often applied for model comparison is the conditional Akaike Information Criteria ($cAIC$), as described by Vaida & Blanchard (2005). This measures the Kullback-Leibler (K-L) divergence between the true model density $f(\cdot|\theta, u)$ and the conditional density of a model candidate $g(y|\hat{\theta}, \hat{u})$. The $cAIC$ is defined as

$$cAIC = -2log(g(y|\hat{\theta}, \hat{u}) + 2K. \tag{2}$$

$log(g(y|\hat{\theta}, \hat{u})$ is the conditional log likelihood, i.e., it is a metric do describe how likely it is that the model $g(\cdot)$ generated the data y. The big difference to the marginal AIC ($mAIC$), which would be preferred in case of a focus on the population parameter, is that the likelihood is also conditioned on the area-level random effects $\hat{u}$ (Vaida & Blanchard, 2005). Since it would ignore the area-level structure, it is not reasonable to use the $mAIC$ if the focus of interest is on the cluster level and not on the the overall population effect (Vaida & Blanchard, 2005).

K is a bias correction term; different versions have been proposed, but due to computational costs and closer approximation under realistic assumptions, K_{MLE} as developed by Vaida & Blanchard (2005) is used for the present term paper, following Lee et al. (2023). The term is defined as

$$K_{MLE} = \frac{N(N-p-1)}{(N-p)(N-p-2)}(\rho+1) + \frac{N(p+1)}{(N-p)(N-p-2)}. \tag{3}$$

The problem is, however, that the proposed model selection criterion is not scale invariant. That means, if the dependent variable y changes, e.g., due to a transformation, the $cAIC$ for different models cannot be compared with each other for variable selection. This is especially vibrant if not a fixed transformation (e.g., a log transformation) is applied but a data driven transformation which changes based on the covariates included. One of those data driven transformations often applied in research is the Box-Cox transformation (Box & Cox, 1964). It contains a parameter λ which can be estimated. In the estimation of λ the goal is to find the λ which maximizes the normal likelihood of the dependent variable (Box & Cox, 1964). For the estimation, typically restricted maximum likelihood estimation (REML) is used (Gurka et al., 2006, 2007). The box-cox transformation is defined as

$$T_\lambda(y_{ij}) = \tilde{y} = \begin{cases} \frac{(y_{ij}+s)^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0 \\ log(y_{ij}+s) & \text{if } \lambda = 0 \end{cases}, i = 1, ..., D \text{ and } j = 1, ..., N_i \tag{4}$$

Instead of the model above, using the transformation the following model is estimated: $T_\lambda(y) = \tilde{y} = X\beta + Zu + \epsilon$. Using this model instead, the goal is to get normally distributed random effects and residuals - two crucial assumptions of the linear mixed model.

To overcome this limitation, one solution is the $JcAIC$ as proposed by Lee et al. (2023). The present term paper aims to outline this approach and investigate, to what extent the usage of the more complex approach improves the statistical analysis. To answer this question, I conducted a simulation study comparing different variable selection approaches under varying data-generation processes (DGP's). In the following, first the approach by Lee et al. (2023) is outlined in detail. Afterwards, I introduce the simulation setup including the compared methods and DGP's. The subsequent results section focuses on four different dimensions: The estimated transformation parameter $\hat{\lambda}$, the $JcAIC$ itself, the selected covariates, and the RMSE, before the results are discussed. More information on the plausibility of the DGP functions and the replication results can be found in the appendix.

2 Theoretical Background

The key idea of Lee et al. (2023) is to scale the $cAIC$ proposed above by the Jacobian matrix in order to account for the difference in scales; correcting with the Jacobian ensures that models with differently transformed response variables $\tilde{y}$ are comparable. The Jacobian of the Box-Cox transformation is defined as

$$J(\lambda, y) = \prod_{i=1}^{D} \prod_{j=1}^{N_i} \frac{\partial \tilde{y}_{ij}}{\partial y_{ij}} = \prod_{i=1}^{D} \prod_{j=1}^{N_i} (y_{ij}+s)^{\lambda-1}. \tag{5}$$

It is visible that the Jacobian in this case is calculated as the product of the main diagonal of a D x N_i matrix. The value of the Jacobian primarily depends on the value of the dependent variable y_{ij}, but also on the estimated λ. With a λ close to one, the Jacobian tends to be equal to one as well (as $x^0 = 1$). Assuming $y_{ij} > 1$, the value of the Jacobian increases with increasing λ. Figure 1 illustrates this relationship.

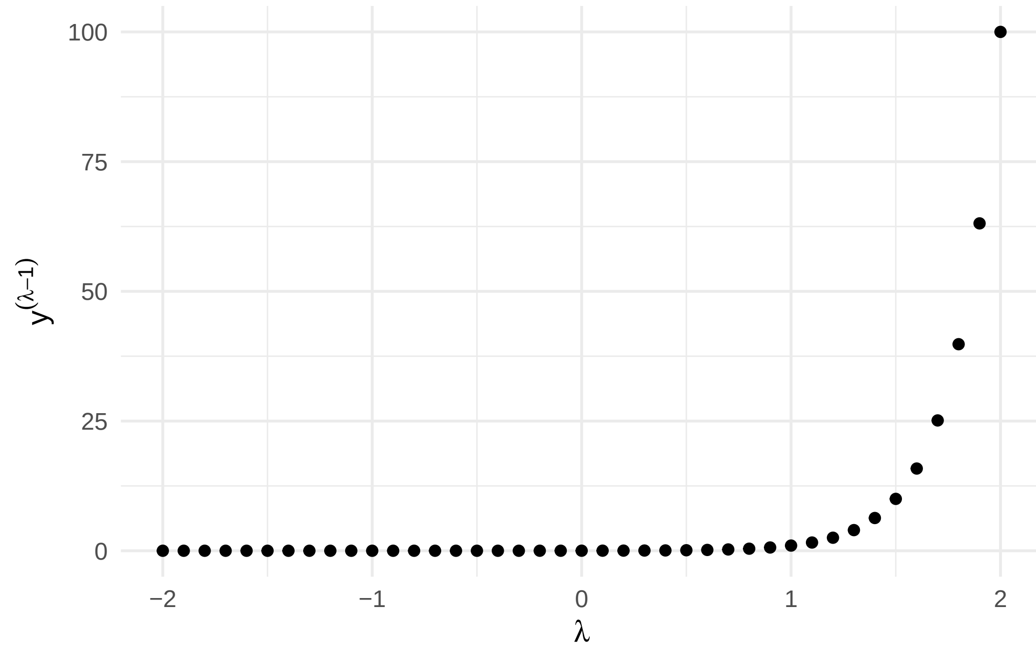


Figure 1: Illustration of relationship between different values of lambda and the calculated value for the respective entry of the Jacobian matrix with fixed y = 100.

Using this definition of the Jacobian and the transformed dependent variable $\tilde{y}$ as well as the conditional true and model densities $f(\cdot)$ and $g(\cdot)$ as introduced above, the true conditional density $h(y|u_0)$ and conditional model density $l(y|\theta, u)$ can be reforumulated as

$$h(y|u_0) = f(\tilde{y}|u_0) \cdot J(\lambda, y) \tag{6}$$

$$l(y|\theta, u) = g(\tilde{y}|\theta, u) \cdot J(\lambda, y). \tag{7}$$

The K-L-distance between those two densities can now be calculated. After some mathematical transformations[1], the final representation of the $JcAIC$ looks like the following:

$$JcAIC = -2log(l(y|\hat{\theta}, \hat{u})) + 2BC = -2log(g(\tilde{y}|\hat{\theta}, \hat{u})) - 2log(J(\hat{\theta}, y)) + 2BC \tag{8}$$

[1]For the calculation steps to arrive at formula (8) please refer to Lee et al. (2023).

For the conditional model density, estimates of θ and u are used. Using formula (7) from above and the estimates yields the first and second expression in formula (8). The $JcAIC$ consists of three terms: $g(\tilde{y}|\hat{\theta}, \hat{u})$ is the conditional model density or conditional likelihood of the transformed dependent variable $\tilde{y}$. $J(\hat{\theta}, y)$ is the Jacobian as introduced above using estimated values for θ. Lastly, BC is a bias correction term as $-2log(l(y|\hat{\theta}, \hat{u}))$ is a biased estimator of $0.5 \cdot JcAIC$. The bias correction term is defined as

$$BC = E_{h(y,u)}[log(l(y|\hat{\theta}, \hat{u}))] - E_{h(y,u)} E_{h(y^*|u)}[log(l(y^*|\hat{\theta}, \hat{u}))] \tag{9}$$

Due to computational complexity, the expectations in the bias correction are estimated using a bootstrap with the following structure:

1. For a model candidate, estimate $\hat{\lambda}$ and transform y to $\tilde{y}$ using $\hat{\lambda}$.
2. Fit the linear mixed model on the transformed response variable $\tilde{y}$. This step yields parameter estimates $\hat{\theta} = \{\hat{\beta}, \hat{G}_0, \hat{\sigma}^2\}$.
3. Based on the estimated parameter of step two, generate $u^{(b)} \sim N(0, \hat{G}_0)$ and $\varepsilon^{(b)} \sim N(0, \hat{\sigma}^2)$. Using those draws, generate a bootstrap sample using $\tilde{y}^{(b)} = X\hat{\beta} + Zu^{(b)} + \varepsilon^{(b)}$.
4. Estimate the bootstrap estimates of the parameters $\hat{\theta}^{(b)}$ and $\hat{u}^{(b)}$ based on the bootstrap sample $\tilde{y}^{(b)}$ from step three.
5. For the second expectation term of the BC, calculate that using $\hat{\theta}^{(b)}$ and $\hat{u}^{(b)}$. The true $\tilde{y}^*$ and y^* are replaced by $\tilde{y}$ and y, respectively.
6. For the first expectation term, back-transform $\tilde{y}^{(b)}$ using $\hat{\lambda}$ to obtain $y^{(b)}$. Re-estimate $\hat{\lambda}^{(b)}$ and re-transform $y^{(b)}$ to obtain $\tilde{y}^{(\hat{\lambda}^{(b)},(b))}$.
7. Refit the model on $\tilde{y}^{(\hat{\lambda}^{(b)},(b))}$ to obtain $\hat{\theta}^{(\hat{\lambda}^{(b)},(b))}$ and $\hat{u}^{(\hat{\lambda}^{(b)},(b))}$.
8. The first expectation term of the BC can now be calculated using $\hat{\theta}^{(\hat{\lambda}^{(b)},(b))}$, $\hat{u}^{(\hat{\lambda}^{(b)},(b))}$, $\tilde{y}^{(\hat{\lambda}^{(b)},(b))}$, $\hat{\lambda}^{(b)}$, and $y^{(b)}$.

Using the estimated values from above, the bootstrap estimate of the bias correction term BC is calculated as

$$BC = \frac{1}{B}\sum_{b=1}^{B}\Big[-\frac{N}{2}log(2\pi\hat{\sigma}^{2^{(\hat{\lambda}^{(b)},(b))}}) - \frac{1}{2\hat{\sigma}^{2^{(\hat{\lambda}^{(b)},(b))}}} \cdot \left(\tilde{y}^{(\hat{\lambda}^{(b)},(b))} - X\hat{\beta}^{(\hat{\lambda}^{(b)},(b))} - Z\hat{u}^{(\hat{\lambda}^{(b)},(b))}\right)^T$$
$$\left(\tilde{y}^{(\hat{\lambda}^{(b)},(b))} - X\hat{\beta}^{(\hat{\lambda}^{(b)},(b))} - Z\hat{u}^{(\hat{\lambda}^{(b)},(b))}\right) + (\hat{\lambda}^{(b)} - 1)\sum_{i=1}^{D}\sum_{j=1}^{N_i} log(y_{ij}^{(b)} + s)\Big]$$
$$-\frac{1}{B}\sum_{b=1}^{B}\Big[-\frac{N}{2}log(2\pi\hat{\sigma}^{2^{(b)}}) - \frac{1}{2\hat{\sigma}^{2^{(b)}}} \cdot \left(\tilde{y} - X\hat{\beta}^{(b)} - Z\hat{u}^{(b)}\right)^T$$
$$\left(\tilde{y} - X\hat{\beta}^{(b)} - Z\hat{u}^{(b)}\right) + (\hat{\lambda} - 1)\sum_{i=1}^{D}\sum_{j=1}^{N_i} log(y_{ij} + s)\Big] \tag{10}$$

It is visible that the term to calculate the bias correction term consists of two averages over the bootstrap iterations B, where each of the terms is the sum of a normal log-likelihood and the logarithm of the Jacobian.

The whole workflow using the measures proposed above can be summarized like the following:

1) Starting point is a full model with all P covariates. Estimate $\hat{\lambda}$ for this, initial optimal, model.
2) For each step $s = 1, ..., S$:
    i) Take into consideration all model candidates with one covariate less than the model before.
    ii) For every candidate, estimate $\hat{\lambda}$ and calculate the $JcAIC$ with the estimated $\hat{\lambda}$ and transformed $\tilde{y}$.
    iii) Choose the model with the lowest $JcAIC$ as optimal model for this step. Start again with i).
3) As soon as no lower $JcAIC$ can be identified in iii), this model is seen as optimal model.

The procedure can also be conducted using a stepwise selection in both directions and not just backward selection as outlined above.

3 Data and Methods

In the present term paper, a simulation study will be conducted. The simulation study follows the same basic design as in Lee et al. (2023), but with slight adjustments as outlined below in more detail[2]. The goal of the simulation study is to compare the proposed solution with

[2] The exact replication of the results by Lee et al. (2023) can be found in Appendix II.

its naive competitors under varying DGP's. Ultimately, the term paper aims to answer the research question, whether the proposed simultaneous selection and estimation outperforms the naive selection in different data scenarios. Before conducting the simulation study and following the results by Lee et al. (2023), I formulate the following hypotheses regarding the results:

- If the dependent variable follows a normal or log distribution, the best results in terms of $JcAIC$ and variable selection are obtained by applying no or a static log transformation, respectively.
- Applying the proposed solution does not yield a considerably worse result than the two natural competitors.
- In the more complex Box-Cox DGP, the proposed solution is expected to outperform the methods using no or a log transformation as well as the naive Box-Cox solution both in terms of the $JcAIC$ and variable selection.
- The models chosen as having the lowest $JcAIC$ also show a low RMSE.
- Filtering out the binary noise variable z_2 is more difficult for all methods than excluding the continuous z_1.

3.1 Data Generation Processes

The four different data generation processes stem from three different distributions. First, as comparison and benchmark setting, a normal distribution for the dependent variable y will be used ("Normal (High)"). As first comparison, a normal distribution with considerably higher variances for the distributions of error terms ε and u will be used ("Normal (Low)"). Normal distributions, however, are not necessarily the most accurate scenario for modeling income data (which is often the target of small area estimation). Therefore, the log distribution will be used ("Log") as well as a box-cox distribution ("Box-Cox"). For all scenarios, the three independent variables are drawn from a normal (x_1, x_3) or binomial (x_2) distribution and both, u and ε are drawn from normal distributions. The data generation processes are summarized in Table 1. The data generation processes for $x_{2,ij}$ $(Bin(1, 0.8))$ and $x_{3,ij}$ $(N(0, 1))$ were the same for all settings.

Table 1: Situations for data generating process, $i = 1, ..., D$ and $j = 1, ..., N_i$

Data setting	y_{ij}	$x_{1,ij}$	μ_i	u_{ij}	e_{ij}
Normal (Low)	$400 - 10x_{1,ij} + 100x_{2,ij} - 10x_{3,ij} + u_i + e_{ij}$	$N(\mu_i, 3^2)$	$U[-3, 3]$	$N(0, 30^2)$	$N(0, 60^2)$

Data setting	y_{ij}	$x_{1,ij}$	μ_i	u_{ij}	e_{ij}
Normal (High)	$400 - 10x_{1,ij} + 100x_{2,ij} - 10x_{3,ij} + u_i + e_{ij}$	$N(\mu_i, 3^2)$	$U[-3, 3]$	$N(0, 10^2)$	$N(0, 20^2)$
Log	$exp\{10 - x_{1,ij} + x_{2,ij} - 0.5x_{3,ij} + u_i + e_{ij}\}$	$N(\mu_i, 2^2)$	$U[2, 3]$	$N(0, 0.4^2)$	$N(0, 0.8^2)$
Box-Cox	$[(10 - x_{1,ij} + x_{2,ij} - 0.5x_{3,ij} + u_i + e_{ij}) \cdot (-0.5) + 1]^{\frac{1}{-0.5}}$	$N(\mu_i, 2^2)$	$U[2, 3]$	$N(0, 0.4^2)$	$N(0, 0.8^2)$

Figure 2 presents histograms of the four data generation processes of the first monte carlo iteration. A full plausibility check for all DGP's can be found in Appendix I.

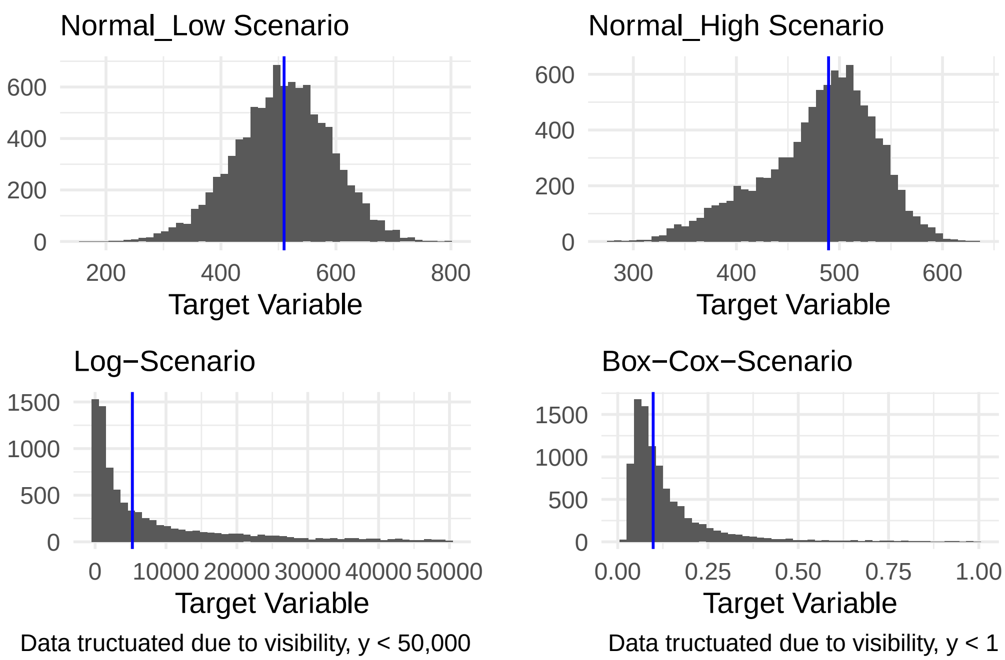


Figure 2: Histogram of the dependent variable for all DGP's. Blue line indicates the median.

Additionally to the three covariates used in the DGP, two irrelevant variables $z_1 \sim N(1, 0.1)$

and $z_2 \sim Bin(1, 0.4)$ are included in the model setup. The simulation study by Lee et al. (2023) only used one additional variable (z_1); the term paper deviates from that since it is assumed that for areas with few observations, correlations will exist by chance and filtering out this variable will be considerably harder for the methods.

For the rest of the setup, the same settings have been applied: Overall, $N = 10,000$ observations were generated in $D = 50$ domains. The sample size per domain was drawn once for all setups and iterations as $n_i \sim Unif(0, 29)$, yielding a slightly higher number of observations as in Lee et al. (2023) but with the same range ($n_i \in [0, 29]$). Observations were sampled using simple random sampling. Due to computational reasons, only $R = 300$ iterations have been conducted with $B = 50$ bootstrap iterations for the bias correction term instead of $R = 500$ and $B = 200$ in the original paper.

3.2 Compared Methods

In total, four different variable selection methods will be compared based on the data generation processes outlined above. First, as a benchmark, a stepwise variable selection based on the $cAIC$ without applying a transformation on the dependent variable y will be conducted ($T_\lambda(y_{ij}) = y_{ij}$, "No" hereafter). The second method is again a stepwise variable selection, but after the dependent variable was log-transformed ($T_\lambda(y_{ij}) = log(y_{ij})$, "Log" hereafter). As third scenario, a data driven box-cox transformation as defined above will be used for transforming y, but after the stepwise variable selection was conducted on the original scale ("Box-Cox Naive" hereafter). Lastly, the process proposed above, stepwise variable selection but based on the $JcAIC$, will be used ("Box-Cox Optimal" hereafter).

Overall, the simulation study is organized in a loop over three levels: R iterations, four DGP's, four methods. In every iteration r, a new population is generated and a new sample is drawn, always with the same sample size per domain. The four methods are then applied all to the same sample. With $R = 300$ iterations, this procedure yields $300 \cdot 4 \cdot 4 = 4,800$ methods. For each method, the final $JcAIC$, the optimal $\hat{\lambda}$ (if applicable), the log-likelihood, the value of the Jacobian, the bias-correction term, the RMSE, and the variables selected for the final model with the lowest $JcAIC$ are stored and used for the analysis later. Those are also the metrics which I will use for the analysis and comparison later.

Additionally, prediction accuracy is evaluated using the RMSE defined as $RMSE = \sqrt{\frac{1}{R}\sum_{r=1}^{R}(\hat{y} - y)^2}$. The model, which has been selected as optimal by the respective method, is used to predict the dependent variable for the whole population from which the sample was drawn earlier. For the sampled domains, the random intercept is included; in case of non-sampled domains, prediction is done on the population level. Before the evaluation, the predictions $\hat{y}$ are back-transformed either using $e^{\hat{y}}$ for the log-method or $(\hat{y} \cdot \hat{\lambda} + 1)^{\frac{1}{\lambda}}$ for the box-cox approaches. Due to Jensen's inequality, a bias is introduced for the log-method through the back-transformation, which is considered to be irrelevant as no content-wise interpretation of the dependent variable is conducted.

For the data generating processes Normal (High) and Normal (Low) it is expected that the first method (stepwise variable selection withtout transformations) will outperform the other approaches as no transformation is necessary prior to the variable selection. $\hat{\lambda}$ is expected to be approximately 1. Regarding the Log scenario, the optimal transformation parameter λ is expected to be approximately 0. Subject of the analysis will be here to what extent the variable selection before the transformation differs from the sequential one proposed by Lee et al. (2023). Lastly, for the box-cox data generation process, the data driven transformation are expected to outperform the static transformations. Regarding the comparison between methods three and four, the analysis will focus both on changes in estimated λ and on the variables selected. Here, $\hat{\lambda}$ is expected to be approximately -0.5.

All reproduction materials can be found on GitHub: https://github.com/nippisch/Replication-Lee-et-al.

4 Results

4.1 Estimation of Lambda

Since the first two methods did not apply a data-driven transformation with an estimated λ, Figure 3 shows the distribution of $\hat{\lambda}$ of the methods Box-Cox (Naive) and (Optimal).

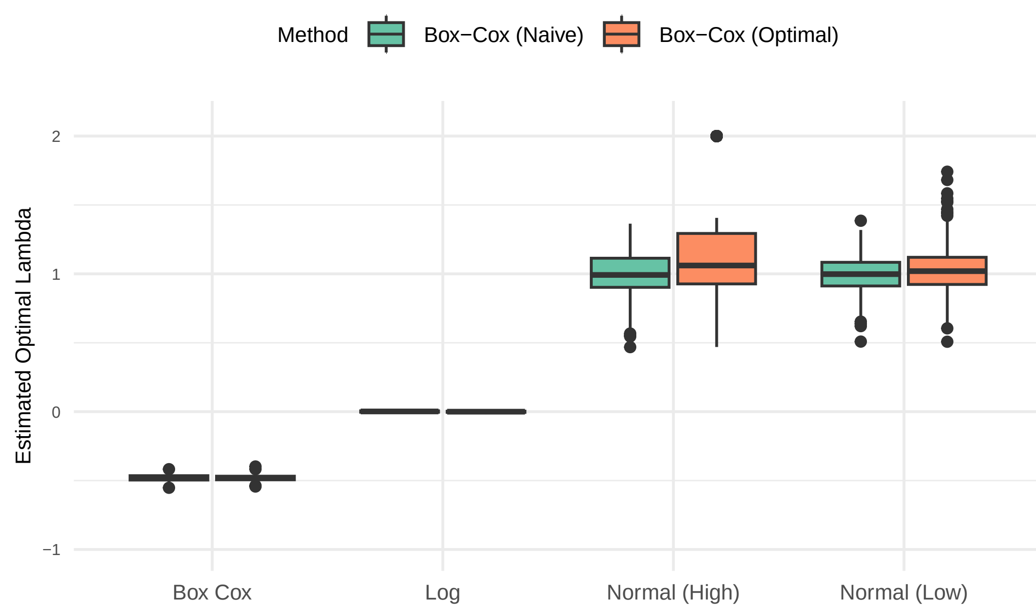


Figure 3: Box-Plot of the estimated optimal lambdas for the Box-Cox Methods per DGP.

For the Box-Cox and Log DGP, the $\hat{\lambda}$ is very close to the expected -0.5 and 1, respectively. In the Normal DGP's, however, more heterogeneity is visible: Even though the median in all four cases is approximately the expected 1, both methods show a considerable variation. In the case of Box-Cox (Optimal) and the Normal (High) setup, some outliers to the top are visible. Those outliers are 67 cases in which $\hat{\lambda}$ was estimated to be 1.9999397. The fact, that all of those $\hat{\lambda}$'s have the same value is likely due to the fact that the possible range of the parameter was set to $\lambda \in [-2, 2]$ and acts as the upper boundary.

Interestingly, those deviations to the top happened to this extent only in the setup, which can be seen as the most optimal for a linear mixed model: A normally distributed dependent variable with low variances for ε and u. One reason for this deviation might be that in this particular setup, the normal distribution is sometimes slightly left-skewed. The left graph in Figure 4, however, shows that even though a considerable amount of too high estimated $\hat{\lambda}$'s is for the most extreme skewed densities, but the picture is not as clear as expected.

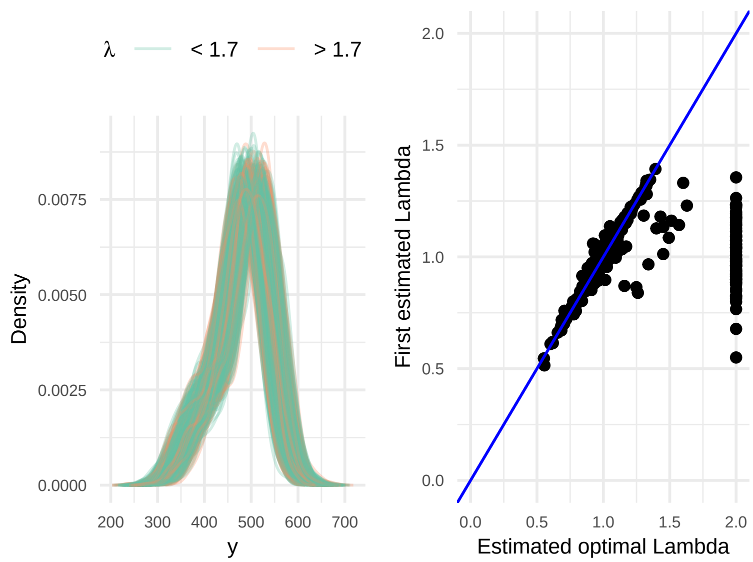


Figure 4: Left: Densities of the dependent variable in the Normal (High) setup. Right: First estimated and optimal Lambda for the Normal (High) setting and Box-Cox (Optimal). Blue line indicates 45°.

A second possible reason for the high estimations might be some kind of path dependency: If the initially estimated $\hat{\lambda}$, which is used for the transformation before the stepwise procedure starts, is exceptionally high, than this might lead to falsely assumed high values subsequently. The right graph of Figure 4 shows that the association between the first and the optimal $\hat{\lambda}$ is very strong, but not for the very high $\hat{\lambda}$. For them, we see varying initial $\hat{\lambda}$ around the optimal value 1, so apparently the hypothesis of path dependency can also be discarded.

Upon further investigation it became clear that the variable x_2 apparently causes this pattern.

Removing x_2 from the model yields a bimodal distribution of residuals as it is a binary predictor with a very high coefficient in the DGP. The problem might be less severe in the Normal (Low) setting as the high variances of random effect and error term might reduce the strength of the pattern. Further investigation revealed that estimating $\hat{\lambda}$ moves towards the upper boundary in all of the $R = 300$ iterations conducted. Interestingly, just in 68 cases the high $\hat{\lambda}$ dominated the $JcAIC$ and let the procedure choose this model. Subsequent iterations of the $JcAIC$ all yielded a $\hat{\lambda} \approx 2$. In the other cases, however, the model without x_2 was not chosen even though $\hat{\lambda} \approx 2$ in those cases as well. Investigating characteristics of the samples did not yield any results: Descriptive statistics like skewness, kurtosis, or range of the samples did not differ based on whether the $\hat{\lambda} \approx 2$ model was chosen or not. Therefore, it can be assumed that it is an artifact of the iterative procedure. Why exactly this behavior occurs just in a few cases, however, remains open to further research.

4.2 JcAIC and its Components

The next natural step for the comparison is to investigate the $JcAIC$ the different methods yielded. Due to the construction of the metric, they are compareable across methods, so the lowest $JcAIC$ per setup should indicate the best fitting model. Figure 5 shows exactly that. It is visible, that for the Box-Cox and Log DGP, applying no transformation yields a considerably higher $JcAIC$ compared to the other methods. The other three, however, show a comparable performance within those settings. The same is true for the Normal DGP's: All methods yield approximately the same $JcAIC$ with a little higher variation for the Box-Cox methods.

The fact that Box-Cox (Optimal) does yield approximately the same median $JcAIC$ as the other three methods even though the $\hat{\lambda}$ partly was very far off is surprising. To further investigate what potential reasons for this finding might be, Figure 6 shows the $JcAIC$ and the decomposition into the three different terms. As reminder, the $JcAIC$ was defined as $JcAIC = -2log(g(\tilde{y}|\hat{\theta}, \hat{u})) - 2log(J(\hat{\theta}, y)) + 2BC$.

It is visible that the $JcAIC$ values do not show a clear pattern, neither between models with a very high $\hat{\lambda}$, nor between models which correctly identified the covariates and not. The expectation would be, however, that wrong models with a high value for $\hat{\lambda}$ have a considerably lower $JcAIC$. The potential reason, why this is not the case, is visible in the graphs of the Log-Likelihood and the Jacobian: The wrong models with high $\hat{\lambda}$ show a considerably lower Log-Likelihood with the high $\hat{\lambda}$ being the more precise predictor for a low Low-Likelihood. In the $JcAIC$, those models would receive a very high value (as the Log-Likelihood (a negative value) is multiplied by -2). This effect, however, gets almost cancelled out by the Jacobian: The same models show a considerably higher value here, bringing the $JcAIC$ down again (as the Jacobian (a positive value) is multiplied by -2 and added to the Log-Likelihood). The bias correction term does not cancel out this effect since wrong models with high $\hat{\lambda}$ show a very low value, but the other models show a high value in comparable absolute size.

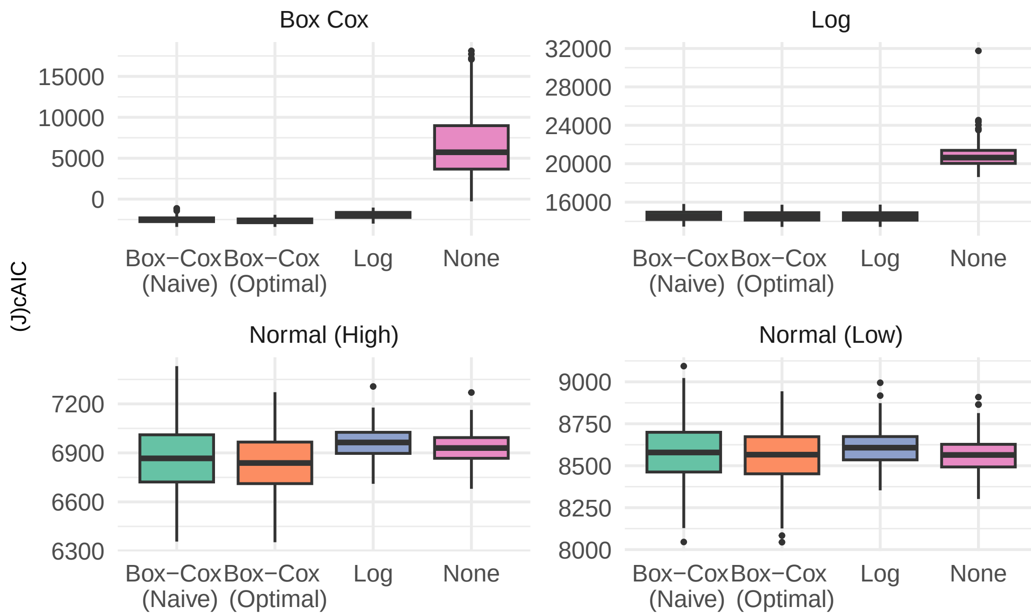


Figure 5: (J)cAIC per DGP and Method. Note: Y-Axis changes from DGP to DGP for better comparability within DGP's.

Table 2: Mean of Log-Likelihood, Correction Term, and JcAIC per correctly identified model and estimated lambda, ascending by JcAIC.

Model	Lambda	-2Log-Likelihood	-2Jacobian + 2BC	JcAIC
False	> 1.7	16920.56	-9956.94	6963.63
Correct	< 1.7	7169.33	317.63	7486.97
False	< 1.7	7939.04	-9.08	7885.32

Table 2 further illustrates this result: The wrong models yield a far lower $JcAIC$ due to the big $\hat{\lambda}$ and the impact on the Jacobian, even though the Log-Likelihood is very low. Correctly identified models show up here after a considerable gap.

To sum up, the results of the $JcAIC$ are mostly as expected, even though the difference between Box-Cox (Optimal) and the other methods is not as big as expected. More surprising is, however, that the construction of the metric apparently prefers models with a high $\hat{\lambda}$, what might disguise bad model fit.

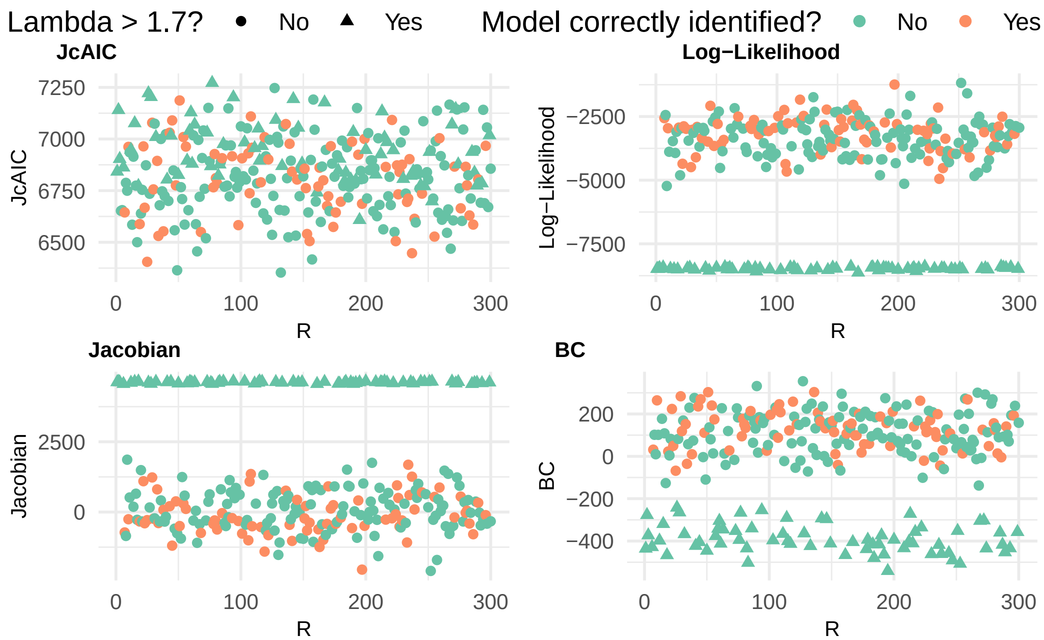


Figure 6: JcAIC and single terms for the Normal (High) DGP and the Box-Cox (Optimal) method.

4.3 Variable Selection

Ultimately, the key objective of the whole procedure is to select optimal covariates. Since the true DGP is known for a simulation study, the methods' ability to select the correct variables can be checked, too. Figure 7 presents the mean percentage of correctly identified models across all $R = 300$ iterations. For the Box-Cox and Log DGP, it is visible that the two naive approaches (no transformation and Box-Cox (Naive)) were able to correctly identify the model in just 17% and 33%, respectively. Box-Cox (Optimal) performed best in the Box-Cox DGP and the Log transformation best in the Log DGP, with slight differences between the two. Over all DGP's, in the Normal (Low) setup the least model were identified correctly. But also in the optimal setting, Normal (High), in just around 75% of the cases the correct variables were selected. What is especially striking is that the Box-Cox (Optimal) performs fairly bad in the two Normal DGP's with just 25% and 16% of correctly identified models.

Those findings point towards a discrepancy between the ability to select the correct variables and the $JcAIC$ ass a metric to assess the quality of a model, as also illustrated by Figure 8. The expectation would be a clear negative, linear trend: A higher percentage of correctly identified models would be associated with a lower mean $JcAIC$. What is visible, however, is no structure at all: Methods with a very low percentage of correctly identified models have approximately the same mean $JcAIC$ as methods with a very high percentage. There is also

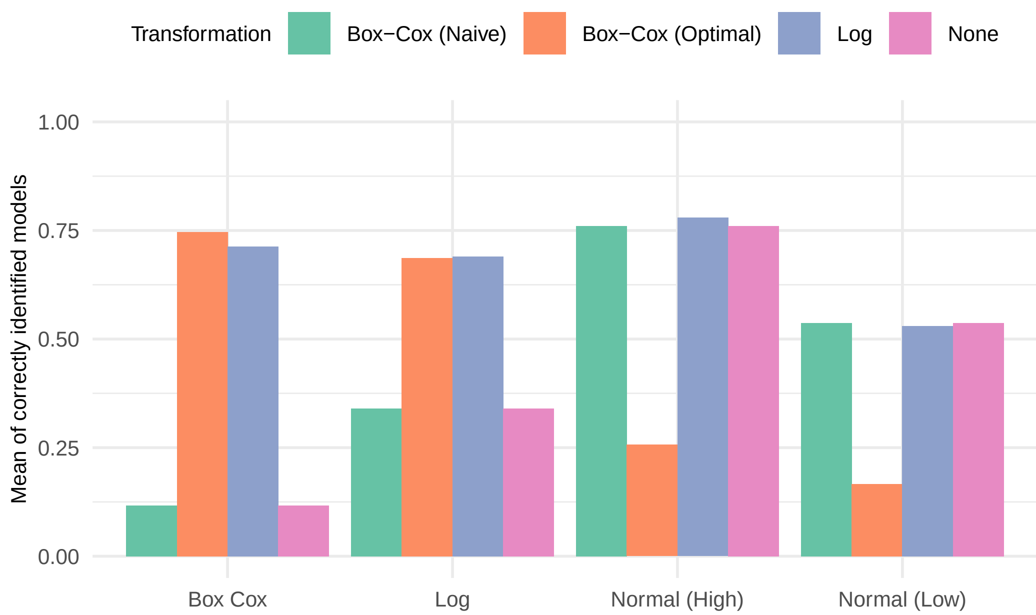


Figure 7: Percentage of correctly identified models per method and DGP.

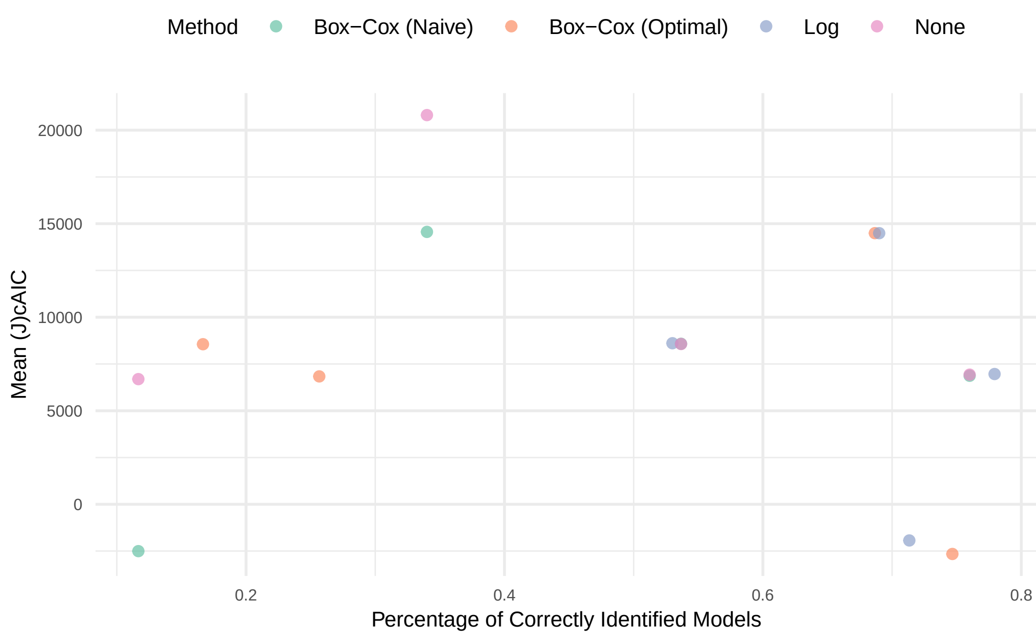


Figure 8: Percentage of correctly identified models and mean (J)cAIC

no clear structure visible regarding the different methods: All methods show a considerable variation both in terms of mean $JcAIC$ and the percentage, with the Log transformation being the most consistent in terms of variable selection accuracy.

Table 3: Percentage of methods with lowest JcAIC per iteration and fraction of those which were correct.

DGP	None	Log	Box-Cox (Naive)	Box-Cox (Optimal)	Correct
Box Cox	0.00	0.00	0.01	0.99	0.75
Log	0.00	0.52	0.00	0.48	0.69
Normal (High)	0.29	0.00	0.05	0.66	0.42
Normal (Low)	0.47	0.02	0.04	0.48	0.35

There is also a second problem in this discrepancy: In practical applications, the model with the lowest $JcAIC$ among model candidates would be considered optimal and used for further steps. In the case of the Normal DGP's, however, this would lead to just a good third of correctly identified models as Table 3 indicates. In each iteration and per DGP, the method yielding the lowest $JcAIC$ was selected. The table now presents, what fraction of those belong to what method. Additionally, the last column indicates whether this (seen as best) model selected the correct covariates. The Box-Cox (Optimal) method is able to yield the lowest $JcAIC$ in the majority of cases, but does not necessarily select the correct variables. Additionally, even though applying no transformation can be seen as the optimal strategy in the Normal DGP's, in two thirds and half of the cases, respectively, Box-Cox (Optimal) has a lower $JcAIC$ with an overall correct variable selection rate of just approximately on third.

Table 4: Percentage of methods with highest Log-Likelihood per iteration and fraction of those which were correct.

DGP	None	Log	Box-Cox (Naive)	Box-Cox (Optimal)	Correct
Box Cox	0.01	0.96	0.01	0.02	0.73
Log	0.00	0.48	0.20	0.32	0.68
Normal (High)	0.02	0.95	0.03	0.00	0.78
Normal (Low)	0.00	0.99	0.00	0.01	0.53

On the contrary, Table 4 presents the same table again but the models were selected based on having the highest Log-Likelihood. Two things are visible: First, Box-Cox (Optimal) gets selected far less than before; mostly, the Log transformation is selected. Additionally, the percentage of correct models among this selection is considerably higher than when using the $JcAIC$ as an indicator.

Lastly, as Table 5 shows, varies the consistency of the variables considerably. Among the wrongly identified models, x_1 is almost always included in the model. Both x_2 and x_3 are selected moderately often by Box-Cox (Optimal) and Log. Against the above specified hypothesis, however, does z_2 not occur more often in the final models than z_1. Interestingly, even though Box-Cox (Optimal) and Log select both x_2 and x_3 more precisely, they also contain z_1 and z_2 more often.

Table 5: Percentage of wrongly identified models containing the respective variables.

Transformation	x1	x2	x3	z1	z2	n
Box-Cox (Naive)	0.99	0.57	0.54	0.26	0.25	674
Box-Cox (Optimal)	0.92	0.85	0.77	0.50	0.52	643
Log	1.00	0.99	0.81	0.42	0.48	384
None	0.99	0.57	0.54	0.26	0.25	674

4.4 RMSE

As Figure 9 illustrates, varies the RMSE behavior considerably. Overall, the RMSE is far higher in the Box-Cox and Log setting compared to the normal DGP's. In the two normal settings the four models behave approximately similar in terms of median RMSE, however, it is visible that the range of possible RMSE's is considerably bigger for the Box-Cox (Optimal) method. The same pattern is not visible in the Log and Box-Cox setting. In the former, the Box-Cox approaches and the Log method perform approximately equal, whereas applying no transformation yields a higher RMSE. In the Box-Cox setting, however, both Box-Cox Methods show a far bigger RMSE than the Log method and applying no transformation, even though both methods assume a wrong distribution of y.

To check whether the $JcAIC$ also captures the prediction quality, Figure 10 shows both the RMSE and $JcAIC$ per DGP. Ideally, the plots would show a linear positive relationship where low RMSE-values are associated with a low $JcAIC$. Such a relationship can be observed, however, weaker than expected and with considerable deviations. Especially the two normal settings show, that the same RMSE can be associated with very different values for the $JcAIC$; Log and Box-Cox DGP show very different RMSE values for the same $JcAIC$. Interestingly, whether the model is correctly specified or not does not matter much for neither the $JcAIC$ nor the RMSE. One reason might be that even models with a low (high) likelihood might be able to produce good (bad) predictions. Therefore, the $JcAIC$ does not seem to be the optimal metric choice in case low (R)MSE is the goal.

5 Discussion

The present term paper aimed to compare different model selection methods using different transformations. Next to simply applying no transformation or a static log transformation, two procedures using a data driven Box-Cox transformation were compared. The "Naive"

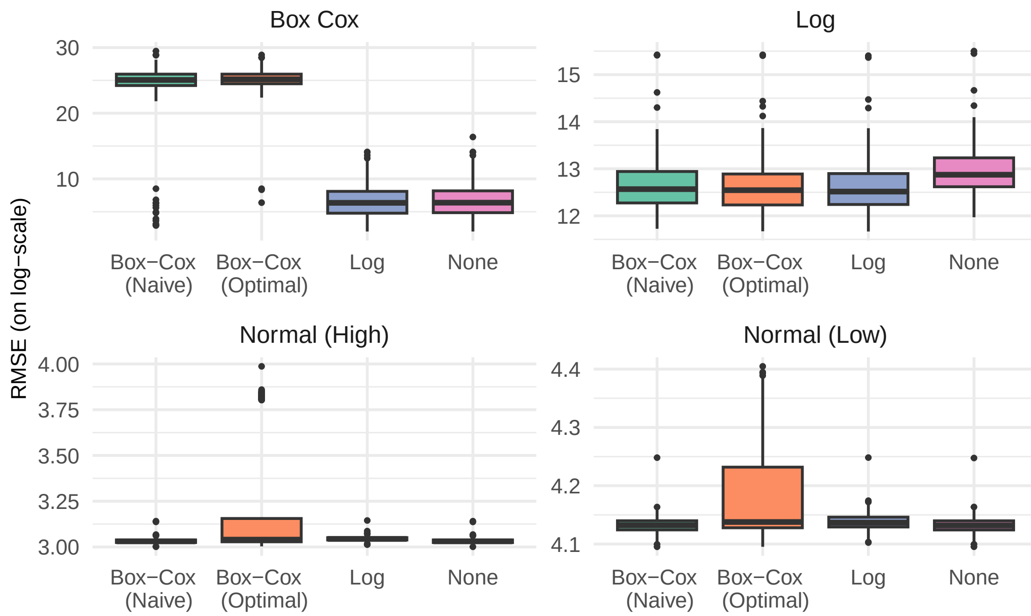


Figure 9: RMSE per DGP and Method. Note: Y-Axis changes from DGP to DGP for better comparability within DGP's.

procedure conducted the variable selection on the original scale and transformed the dependent variable once after the selection, whereas the "Optimal" setup, as proposed by Lee et al. (2023), simultaneously estimates the $JcAIC$ and conducts variable selection based on that. Using this with the Jacobian scaled conditional Akaike Information Criteria allows for a model comparison under varying dependent variables as it is scale invariant. The procedures were compared in four DGP's, namely normally distributed dependent variables with low and high information content, a log distribution, and a box-cox distribution.

Results reveal that the Box-Cox (Optimal) is in most cases able to estimate $\hat{\lambda}$ as expected with some outliers to the top in the Normal (High) setup. Further investigating those revealed that the $JcAIC$, due to its construction as a sum of the Log-Likelihood and the Jacobian, tends to prefer models with a high estimated $\hat{\lambda}$: Those models have a high value for the Jacobian, cancelling out a bad Log-Likelihood which might indicate a bad model fit. Furthermore, investigating the variable selection accuracy showed that the proposed solution is not necessarily able to select the correct covariates, even though it has the lowest $JcAIC$ in many cases. The RMSE-evaluation also indicates a moderate performance at best.

Overall, the results indicate a possible construction problem: The Jacobian disguises possibly bad Log-Likelihoods and gives preference to high $\hat{\lambda}$, ignoring that the estimation of $\hat{\lambda}$ might have gone wrong. In real life applications where the true DGP is unknown, it might be hard to assess whether the $JcAIC$ is low due to good model fit or due to a high $\hat{\lambda}$.

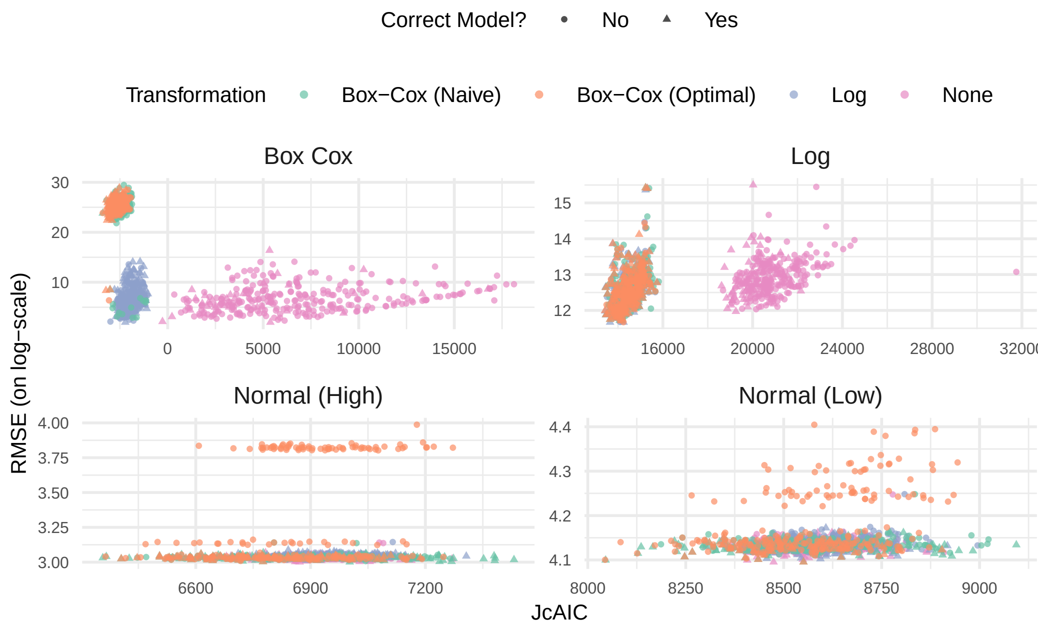


Figure 10: RMSE and JcAIC per DGP. Note: Y-Axis changes from DGP to DGP for better comparability within DGP's.

The results of the term paper have to be interpreted in context of some limitations. First, as Figure 2 indicated, are the distributions of the Normal (High) setup, where the method performed worst, slightly left skewed. This could not be resolved even though the exact same settings have been applied for the DGP. Additionally, there were some slight deviations from the setup in Lee et al. (2023), which might also interact with the findings; that is especially the case for the lower number of bootstrap iterations B (50 instead of 200). As Appendix II shows, however, can the same pattern regarding the different terms of the $JcAIC$ also be found in the case of an almost exact replication.

Further research is necessary to tackle the metrics' limitations outlined in the term paper. With correcting for the scale invariance and enabeling simultaneous estimation of a model selection criterion while applying a data-driven transformation, the $JcAIC$ fills an important gap. To date, however, the preference for highly estimated $\hat{\lambda}$ make it hard to apply it in practice. One promising avenue would be to further investigate the bias correction term: Currently, it only corrects for the biased estimation of the whole criterion without being strong enough to reverse the unintendend preference towards high $\hat{\lambda}$'s. Further corrections like that need to ensure at the same time that no contrary effect is happening, punishing those (in real applications maybe correctly high estimated $\hat{\lambda}$'s) too hard.

6 Appendix I: Plausibility Checks for Data Generation Processes

The goal of this appendix is to demonstrate the generation of a plausible data set, which will serve as base for the simulation studies conducted in the term paper. The document is inspired by the plausibility check documents supplemented for the paper by Rojas-Perilla et al. (2020).

In the following, data will be generated with the same code as used for the term paper. The only adjustment to the code used there is the saving of the population data, which is skipped in the simulation study.

In Figure 11, densities of the different DGP's each for the sample and population are visible. The graphs show a good fit between the population and the sample.

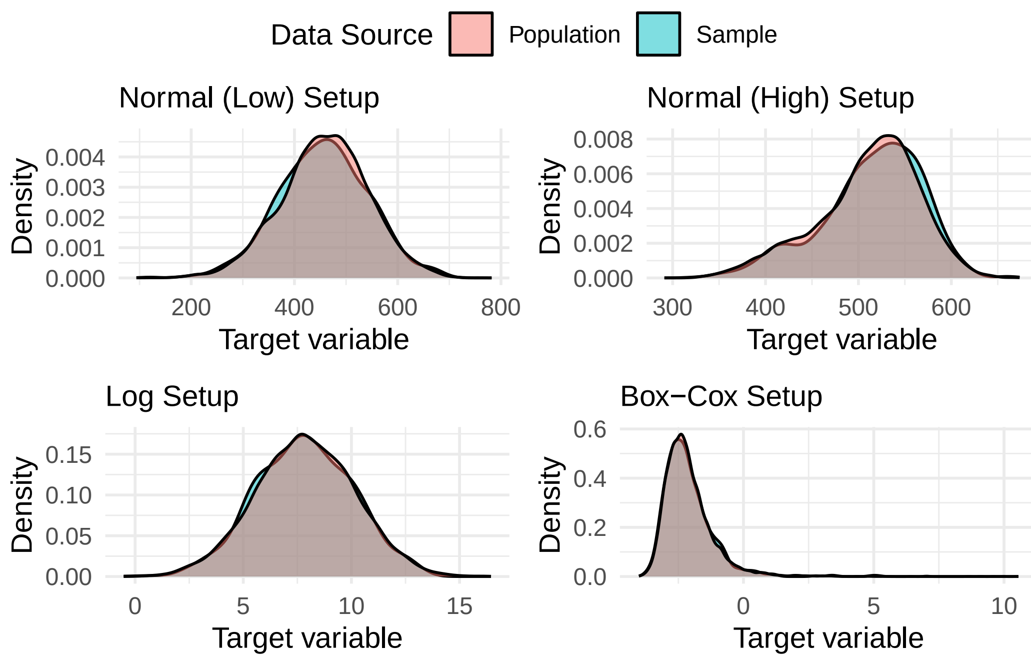


Figure 11: Density of the dependent variable in sample and population for each DGP. Note: For better visibility, a the dependent variable is on log-scale for Log and Box-Cox DGP.

Table 6 shows the mean and median value for each DGP. The difference of the mean and similarity of the median of sample and population for the Log and Box-Cox DGP indicate that some outliers were not included in the sample. Since the outliers were very extreme while unfrequent, this could be expected. In $R = 300$ iterations, as conducted in the simulation study, it is assumed that in some iterations the most extreme outliers were also included.

Table 6: Mean and median values for each DGP and per sample and population.

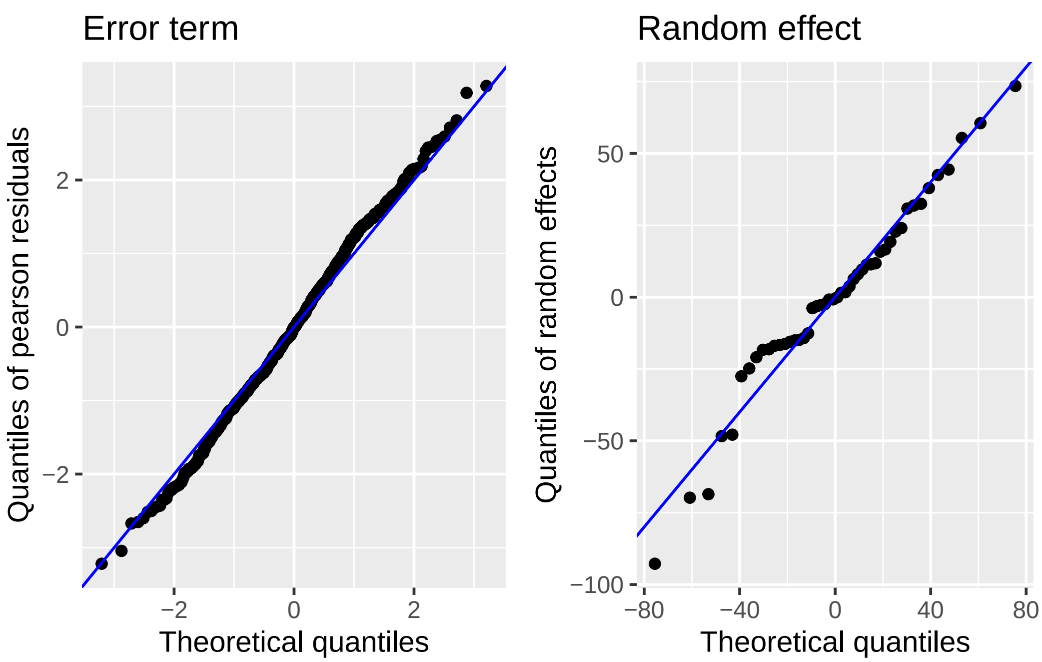
DGP	Source	Mean	Median
Box-Cox	population	9.74	0.10
Box-Cox	sample	0.70	0.10
Log	population	34721.59	2570.19
Log	sample	20657.20	2483.41
Normal (High)	population	510.32	518.41
Normal (High)	sample	514.90	522.85
Normal (Low)	population	459.35	461.43
Normal (Low)	sample	456.07	457.64

In the following, the output of the EBP, exemplary for the Normal (High) DGP, with the true covariates is visible without any noticeable problems. The other DGP's show approximately the same resuts.

```
Empirical Best Prediction

Call:
 ebp(fixed = y ~ x1 + x2 + x3, pop_data = pop_norm_low, pop_domains = "domain",
    smp_data = sample_norm_high, smp_domains = "domain", L = 50,
    transformation = "box.cox")

Out-of-sample domains:  2
In-sample domains:  48

Sample sizes:
Units in sample:  747
Units in population:  10000
                   Min. 1st Qu. Median     Mean 3rd Qu. Max.
Sample_domains        1      10   14.5  15.5625   22.25   30
Population_domains  169     194  199.0 200.0000  207.25  230

Explanatory measures:
 Marginal_R2 Conditional_R2
   0.8337845      0.8645277

Residual diagnostics:
                 Skewness Kurtosis Shapiro_W Shapiro_p
Error         0.003409555 2.666545 0.9972705 0.2483060
Random_effect -0.381954817 3.767777 0.9693479 0.2389101
```

```
ICC:  0.1849601

Transformation:
 Transformation Method Optimal_lambda Shift_parameter
        box.cox   reml       1.225287               0
```

Lastly, diagnostic plots for the same EBP model are visible. QQ-Plots look moderately good with some issues for the extrem values. As the output already indicated, the density plots for the residuals and random effects do not look optimal, but still inside an acceptable range. The plot with Cook's distances shows a few noticeable observations which might be worth investigating in case of a continued working with the model. Lastly, the Log-Likelihood plot in terms of λ shows a maximum very close to the desired 1 in the case of the Normal DGP.

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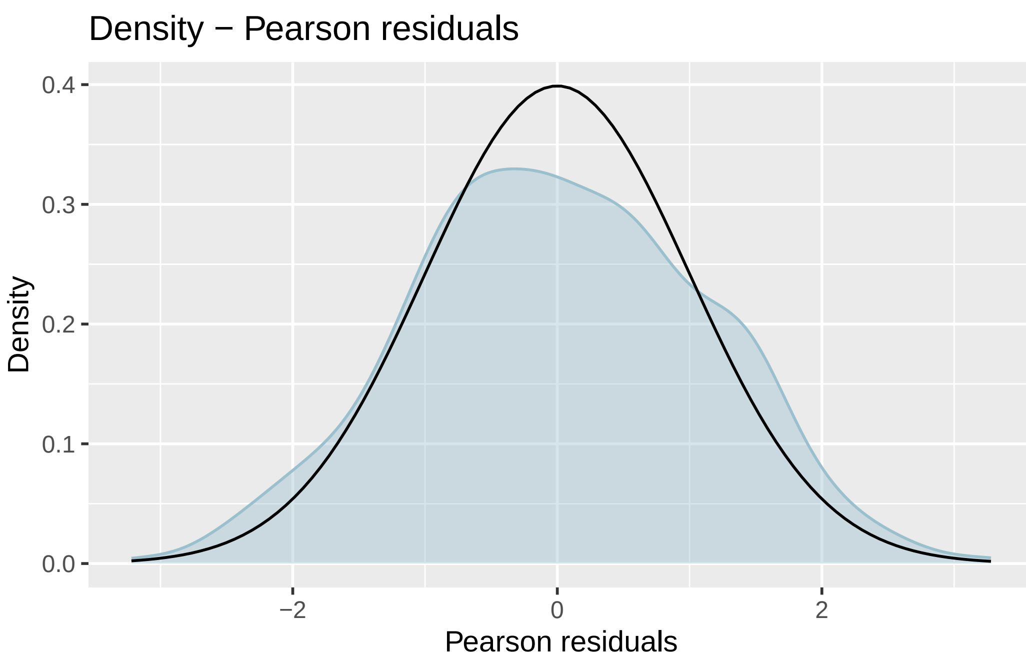

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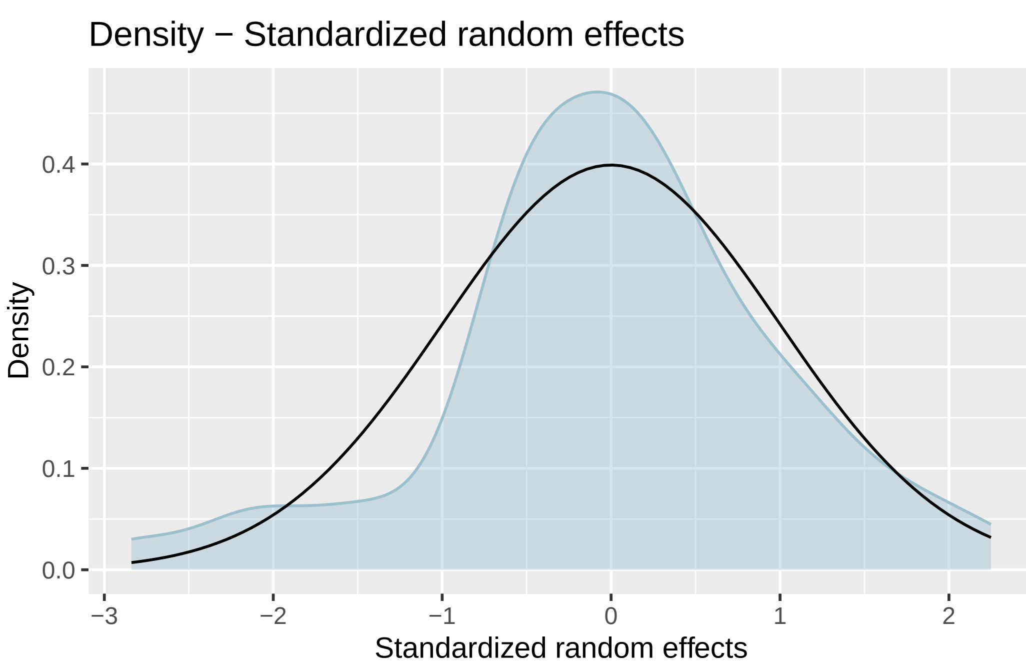

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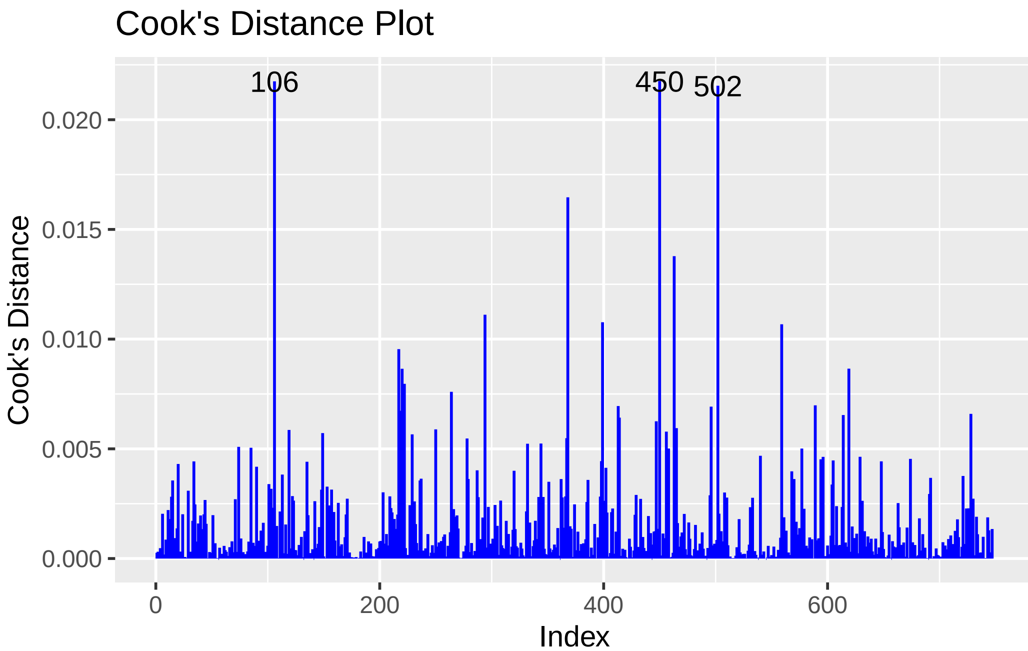


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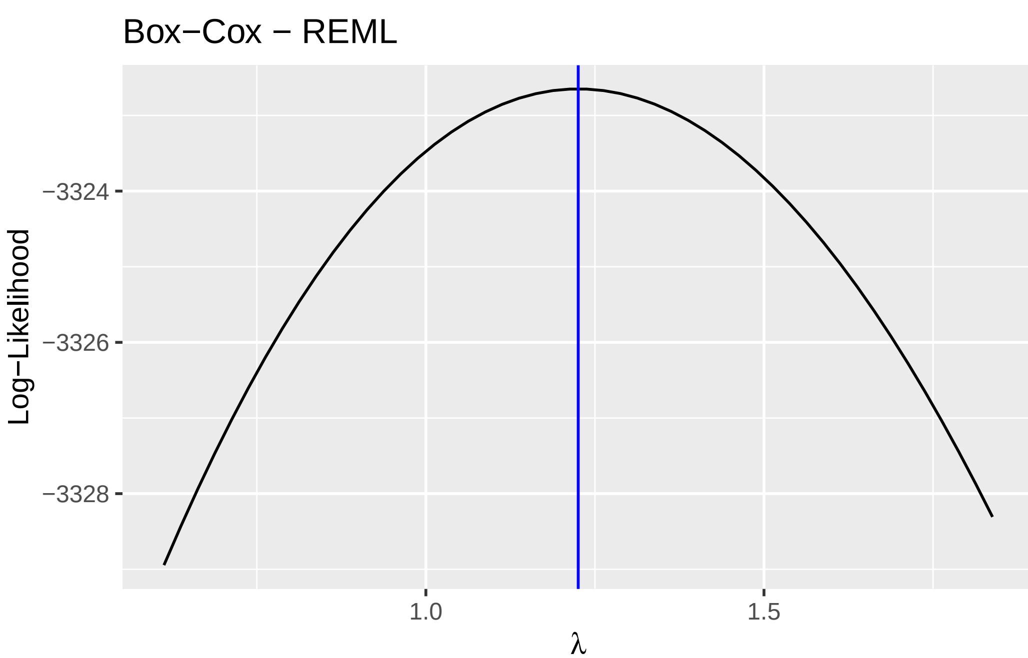

7 Appendix II: Exact replication of results by Lee et al. (2023)

To check whether the results regarding the decomposition of the $JcAIC$ were just an artifact of the DGP's or the low number of bootstrap replications ($B = 50$ instead of $B = 200$), I conducted the simulation study again with the same parameters as in Lee et al. (2023). Since one iteration took about one hour, I used parallel computing on different kernels on five PC's. Still, just $R = 200$ iterations were conducted. Table 7 shows the summary statistics of the optimal $\hat{\lambda}$. Most of the values match the ones of Lee et al. (2023) almost perfectly, so a successful replication can be assumed for the Log, Box-Cox, and Normal (Low) DGP. Regarding the Normal (High) setup, however, it is visible that the present replication contains values for $\hat{\lambda}$, which are not present in the original paper. The fact that the replication does work in other DGP's and for another method (Box-Cox (Naive)) in the Normal (High) setup suggest that it is not simply a false implementation but a interaction of the method and DGP. Further investigation revealed the same pattern as in the simulation study above. The last major deviation from the paper was the unclear sample size per domain. After contacting the author, the original sample sizes could be obtained. Interestingly, the results still deviate from the paper considerably in the Normal (High) setting. However, alternative explanations might also be possible and the reason, why this pattern occured in the present term paper but not in the paper by Lee et al. (2023) could not be answered.

Table 7: Summary statistics of optimal lambda in the replication study.

DGP	Transformation	Min	Q1	Median	Mean	Q3	Max
Box Cox	Box-Cox (Naive)	-0.547	-0.496	-0.481	-0.481	-0.467	-0.428
Box Cox	Box-Cox (Optimal)	-0.552	-0.496	-0.482	-0.482	-0.470	-0.436
Log	Box-Cox (Naive)	-0.021	-0.005	0.001	0.001	0.006	0.027
Log	Box-Cox (Optimal)	-0.025	-0.005	-0.001	0.000	0.005	0.022
Normal (High)	Box-Cox (Naive)	0.490	0.900	0.989	0.995	1.108	1.355
Normal (High)	Box-Cox (Optimal)	0.490	0.935	1.059	1.251	2.000	2.000
Normal (Low)	Box-Cox (Naive)	0.628	0.915	0.986	0.997	1.076	1.353
Normal (Low)	Box-Cox (Optimal)	0.628	0.915	0.987	0.999	1.083	1.353

Even though this discrepancy exists, the same graphs as above for the decomposed $JcAIC$ are visible in Figure 12 and the pattern is even more striking. Again, a very low Log-Likelihood for the models with a very high estimated $\hat{\lambda}$ is visible. Contrary to the graph above, however, the low Log-Likelihood applies almost exclusively for the wrongly identified models. Again, the value of the Jacobian is far higher for high $\hat{\lambda}$ models, cancelling out the effect of the worse Log-Likelihood and yielding a mixed $JcAIC$ for both correct/wrong and high/correct $\hat{\lambda}$ models.

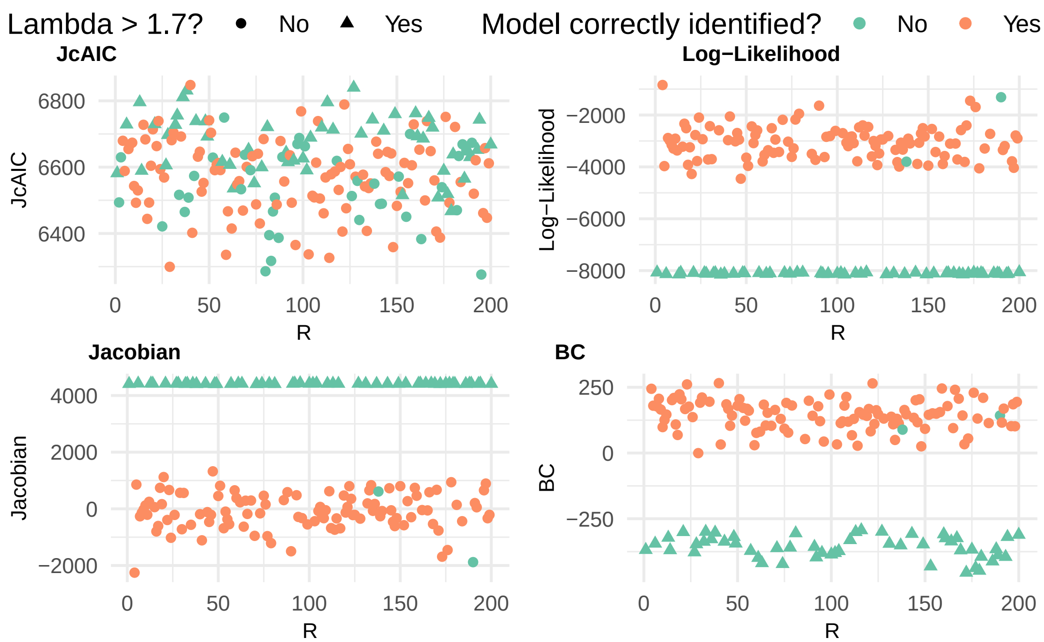

Figure 12: JcAIC and single terms for the Normal (High) DGP and the Box-Cox (Optimal) method in the replication study.

To further investigate the behavior of $\hat{\lambda}$, another simulation run has been conducted with a bigger interval for λ, i.e., $\lambda \in [-4, 4]$. Table 8 shows that estimation of $\hat{\lambda}$ does not simply wanders towards the upper bound but reaches it's maximum at approximately 3.5. The resulting pattern regarding the $JcAIC$ is even more extreme now, as Figure 13 shows, such that the models with the high $\hat{\lambda}$ now have far lower values for the $JcAIC$. The question why $\hat{\lambda}$ this time does not increase over 3.5 also remains open for further research.

Table 8: Summary statistics of optimal lambda in the replication study with a bigger interval for lambda.

DGP	Transformation	Min	Q1	Median	Mean	Q3	Max
Normal (High)	Box-Cox (Optimal)	0.490	0.941	2.957	2.156	3.136	3.499
Normal (Low)	Box-Cox (Optimal)	0.628	0.915	0.987	0.999	1.083	1.353

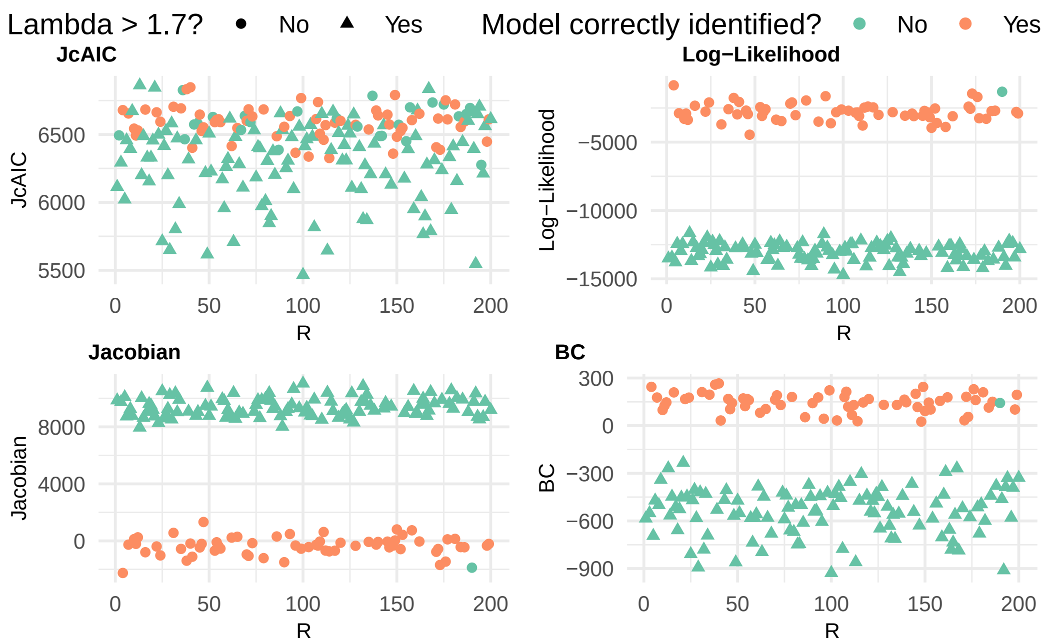

Figure 13: JcAIC and single terms for the Normal (High) DGP and the Box-Cox (Optimal) method in the replication study with a bigger interval for lambda.

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